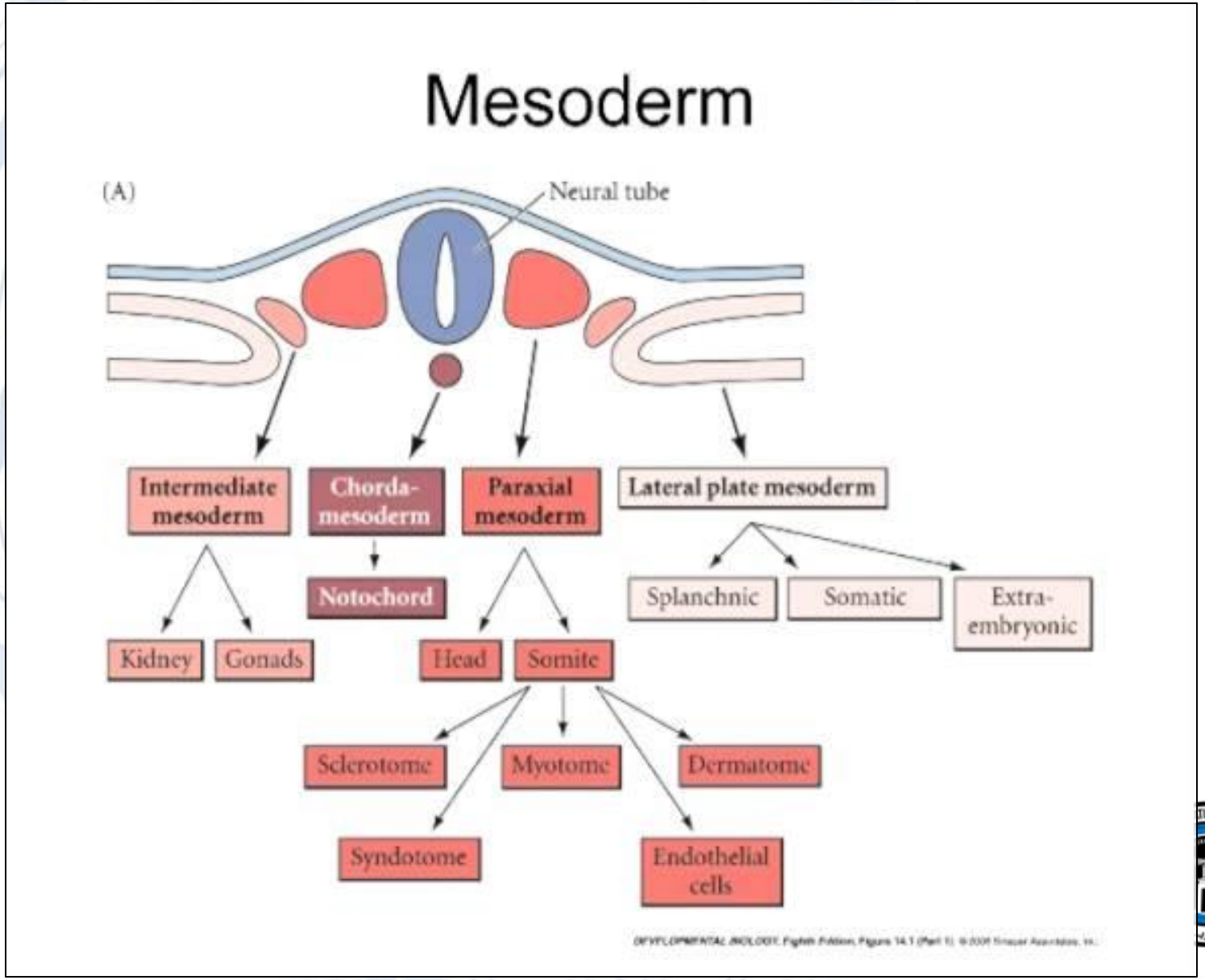
2- Development of the intraembryonic mesoderm



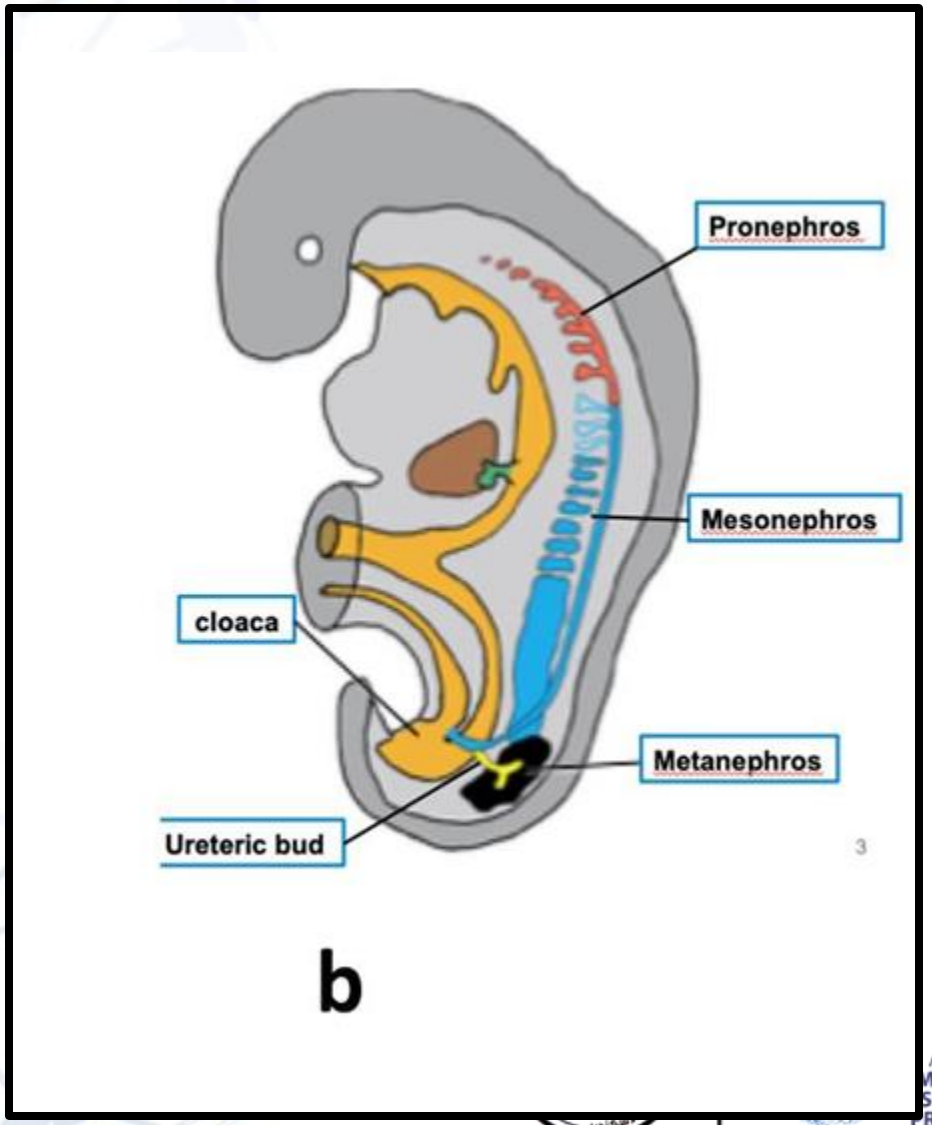
Development of the kidney

- The kidney develops from the intermediate cell mass of the mesoderm.
- In the cervical part, it is segmented into groups of cells called ***nephrotomes.***
- In the caudal part, it gives a longitudinal ridge called ***nephrogenic cord.***
- Three successive sets of kidneys develop during intrauterine life:

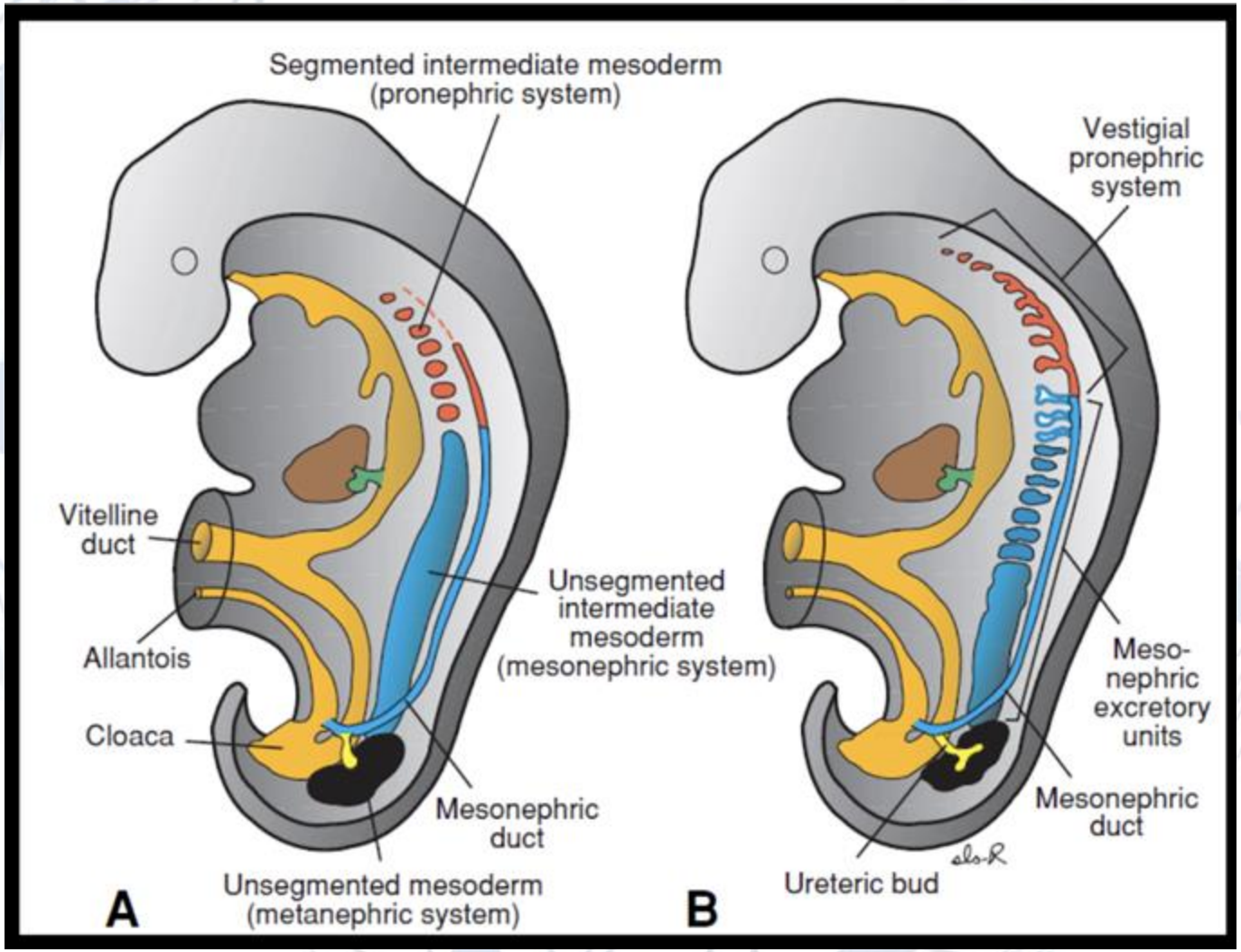
1. Pronephros

2. Mesonephros

3. Metanephros



Development of the kidney



Development of the kidney

1. Pronephros:

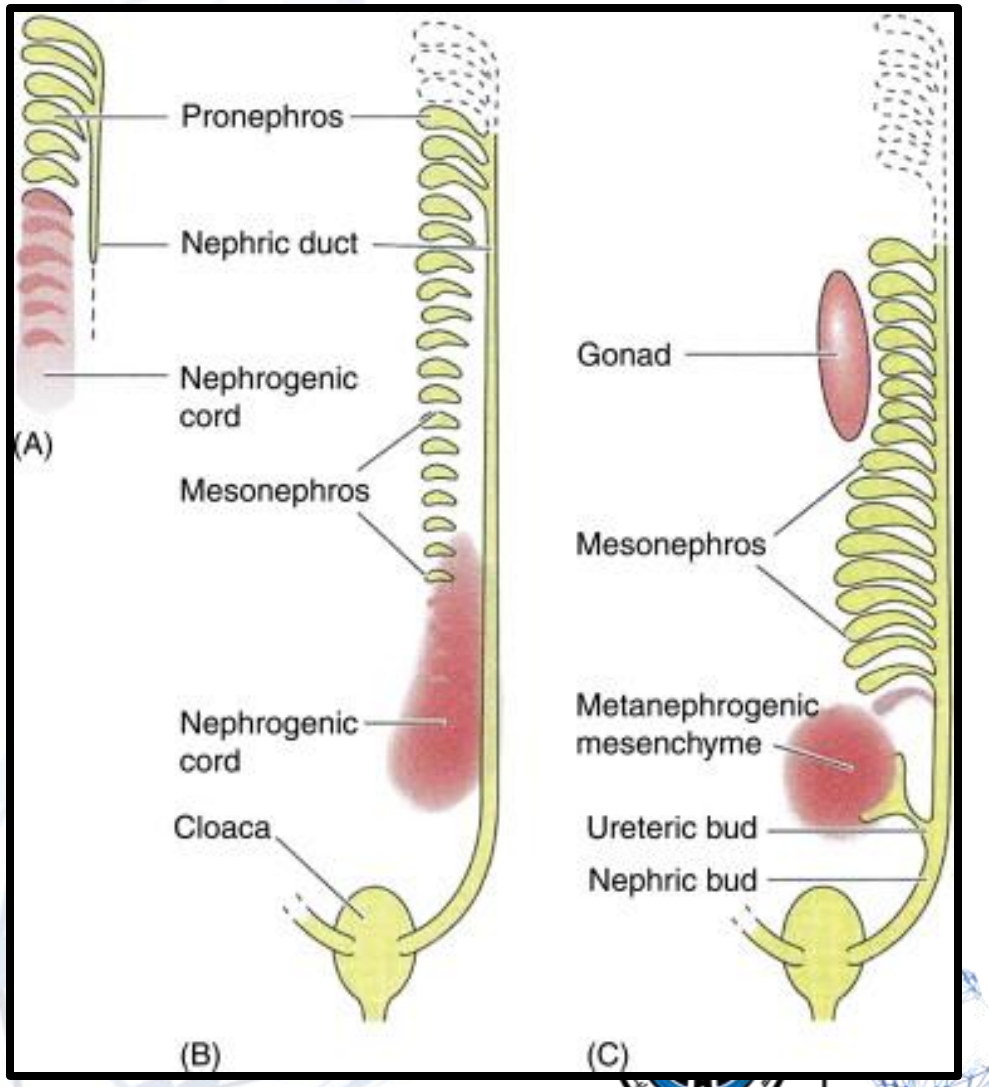
- It is the most primitive Kidney.

- It is a non-functioning kidney.

Site: in the cervical region.

Structure:

- Cervical nephrotomes give 5-7 pairs of pronephric tubules which open in a duct called pronephric duct.
- The tubules degenerate and the duct persist as the mesonephric duct.



ANU MEDICAL SCHOOL PROGRAM

Development of the kidney

2. Mesonephros:

It is the 2nd kidney to appear.

Site: the thoracic and upper lumbar regions.

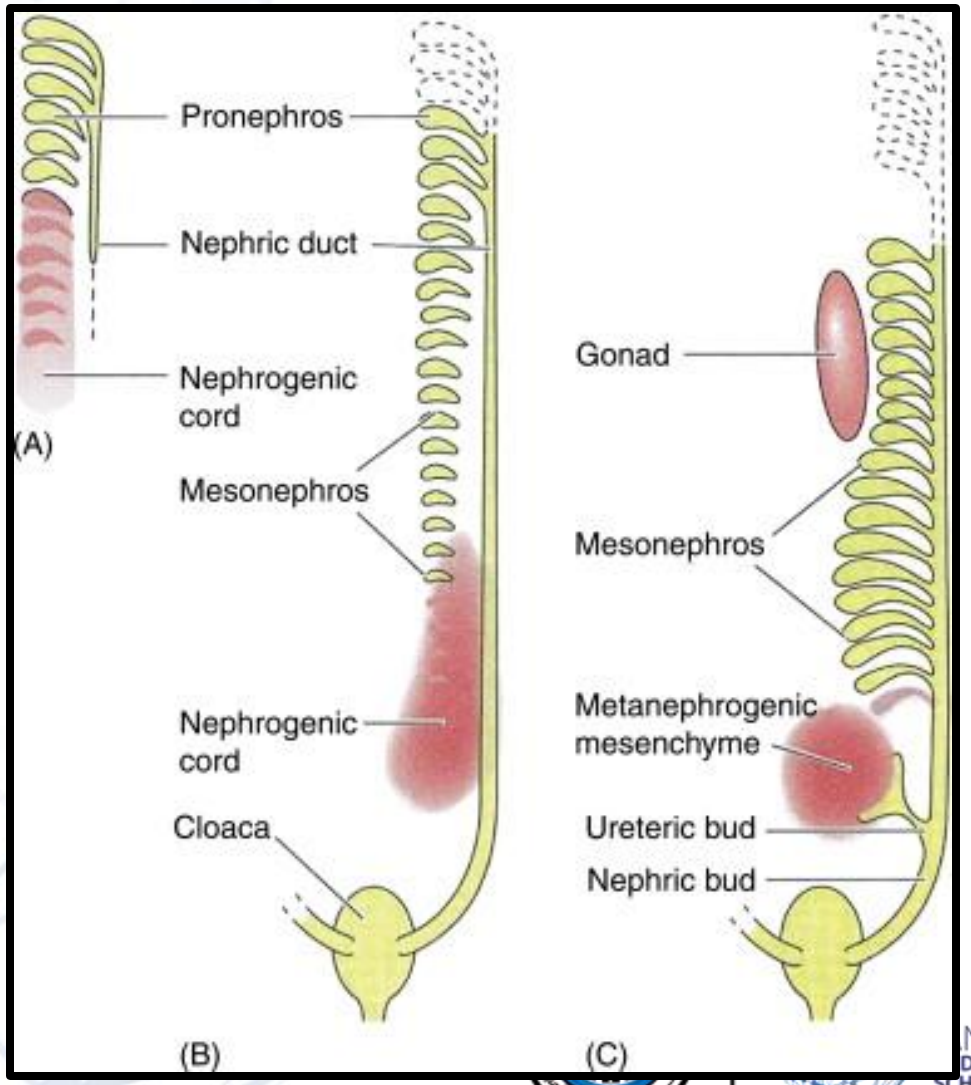
Structure:

- It is formed of about 80 mesonephric tubules.

- Each mesonephric tubule has 2 ends;

  - One end opens in the mesonephric duct
  - The other end differentiates into a cup-shaped bowman's capsule that is invaginated by capillaries forming the Glomerulus.

- The mesonephric duct opens in the cloaca.



Development of the kidney

Fate:

- Mesonephric kidney functions for some time and then degenerates.

- Some mesonephric tubules persist and give efferent ductules in males and epophoron and parophoron in females.

- The mesonephric duct persists as the epididymis, vas deferens, seminal vesicle and ejaculatory duct in males while in females it gives Gartner's duct.

